

2025-2026 – Econ 0039 – Advanced Macro

Lecture 4: The Liquidity Trap

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Disclaimer

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0. Introduction

- ▶ What is the Liquidity Trap? A situation in which :
 - × the nominal interest rate i is zero,
 - × the economy is in a recession,
 - × monetary policy is ineffective because it cannot reduce i any further
- ▶ Why is there a “Zero Lower Bound” for i ? Because i cannot be negative on money: a £1 coin today cannot become a £0.98 coin in one year.
- ▶ Negative i is possible on bonds (i is a contractual arrangement)
- ▶ But then people will hold only money, not bonds, because the nominal interest rate on money is zero
- ▶ Remark: Is it *technically* possible to make the nominal interest rate on money negative? Yes.
 - × *melting money*
 - × A tax on checking accounts if there is only electronic money

0. Introduction

- ▶ We first briefly review the liquidity trap in IS-LM
- ▶ Then we look at facts to show that it is not a theoretical curiosity:
 - × the US in the 1930's (Great Depression),
 - × Japan in the 1990's (Lost Decade),
 - × the US and the UK after 2008 (Great Recession).
- ▶ Then we construct a micro-founded model that gives a rigorous treatment of the liquidity trap.
- ▶ We will see in particular that expectations are crucial to understand what is going on.

1. IS-LM

Assumptions

- ▶ Static model
- ▶ Closed economy
- ▶ Three markets :
 - × goods
 - × money
 - × Bonds
- ▶ Prices are fixed
- ▶ Output is determined by demand
- ▶ Economic agents consume, invest and make portfolio decisions and decide to hold money and bonds.

1. IS-LM

IS

- ▶ Consumption function $C = C(Y - T)$, $1 > C' > 0$
- ▶ Investment function $I = I(i)$, $I' < 0$
- ▶ $i = r + \pi^e$, where π^e is expected inflation ($\pi^e = 0$ because prices are assumed to be fixed)
- ▶ G is given
- ▶ $G + B^G = T + M/P$.
- ▶ G , T , M and B^G given, and satisfy the Gvt budget constraint
- ▶ Equilibrium for income and expenditures: Income Y creates demand that is equal to income:

$$C + I + G = Y \quad (IS)$$

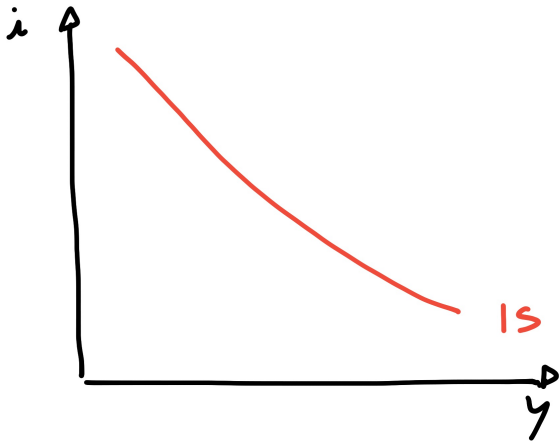
- ▶ IS is downward sloping in the plane (Y, i)

$$C(Y - T) + I(i) + G = Y \quad (IS)$$

- ▶ Fully differentiating wrt Y and i , keeping fixed G and T : $di = \frac{1-C'}{I'} dY$

1. IS-LM

Figure 1: The IS curve



1. IS-LM

LM

- ▶ Money supply is given M/P
- ▶ Real money demand is $L = L(Y, i)$ with $L'_Y > 0$ and $L'_i < 0$
- ▶ $L(Y, i)$ is derived from an arbitrage between money and bonds:
 - × when income is high, money is needed for transaction, so that economic agents use more money (and less bonds) as a mean of savings,
 - × when i is high, the opportunity cost of holding money is high, so that agents hold less money (and more bonds)
 - × Equilibrium (on the money market):

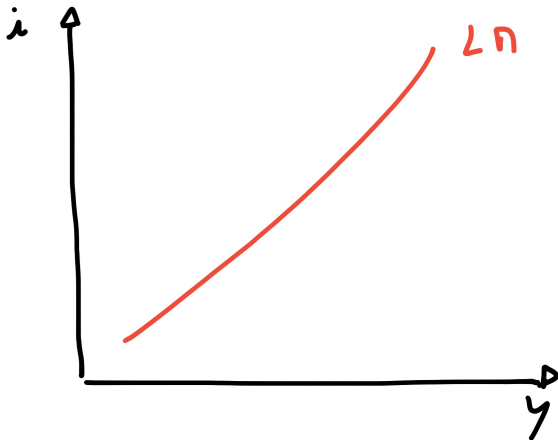
$$L(Y, i) = M/P \quad (LM)$$

- × LM is upward sloping in the plane (Y, i)
- × Fully differentiating wrt Y and i , keeping fixed M/P :

$$di = -\frac{L'_Y}{L'_i} dY$$

1. IS-LM

Figure 2: The LM curve



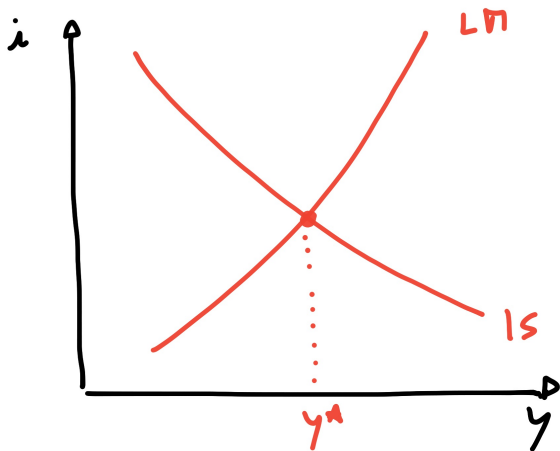
IS-LM

Equilibrium

- ▶ Equilibrium is (Y^*, i^*) such that
 - × Income generates a demand for goods equal to income
 - × Money market clears
 - × By Walras law, the bond market also clears
- ▶ The equilibrium is at the intersection of IS and LM

1. IS-LM

Figure 3: Equilibrium in IS-LM



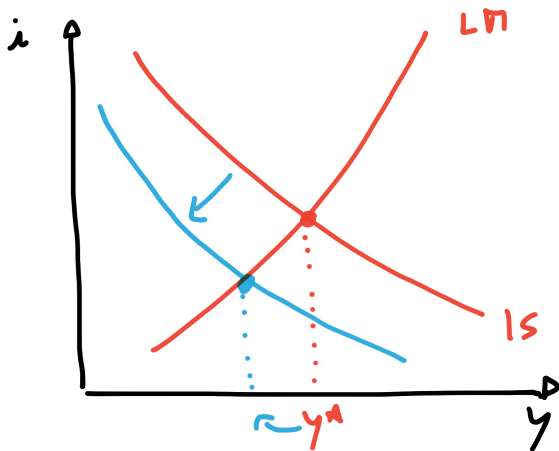
IS-LM

Negative demand shock and the role of expansionary policy

- ▶ Suppose that aggregate demand goes down
- ▶ For example investment goes down following a stock market crash such that the new investment function is $\hat{I}(i) < I(i) \forall i$.
- ▶ This is likely to create a recession
- ▶ Expansionary monetary policy can restore a high Y equilibrium:
 - × Expansionary monetary policy (associated with debt reduction $dM/P = dB^G$) reduces i
 - × a drop in i increases I

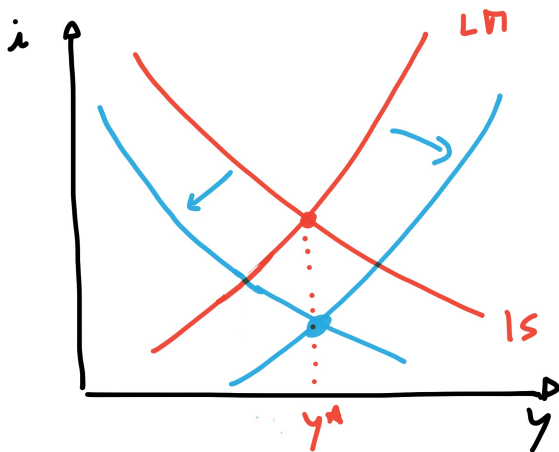
1. IS-LM

Figure 4: A negative shock to aggregate demand



1. IS-LM

Figure 5: Expansionary monetary policy



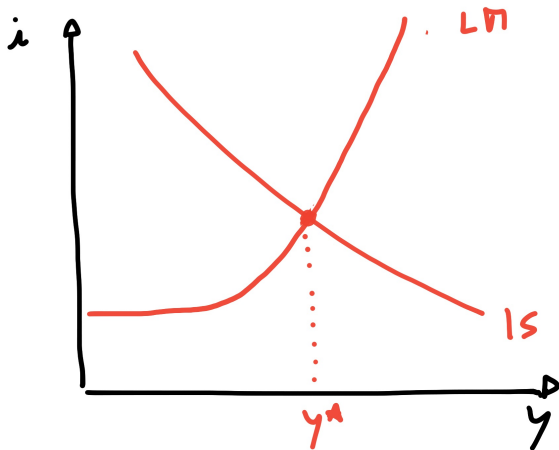
IS-LM

The liquidity trap

- ▶ Assume there is a level of i that is so low (say $i \approx 0$) such that
 - × the demand for bonds is very low (because the return is almost zero)
 - × an increase in money supply cannot decrease any more i because all the money distributed is kept in cash and is not used to buy bonds
- ▶ In the IS-LM language, $L'_i \rightarrow -\infty$ when i is at the *Zero Lower Bound*.
- ▶ Monetary policy cannot reduce i , and is therefore ineffective.
- ▶ If the government cannot run a deficit (because public debt is already high), then there is no possibility to eliminate the recessionary effect of a negative demand shock.

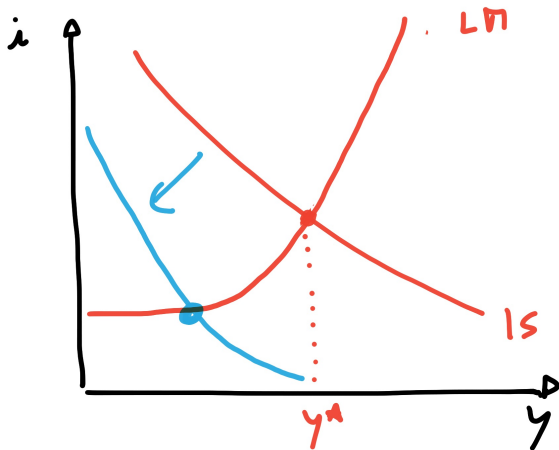
1. IS-LM

Figure 6: The liquidity trap



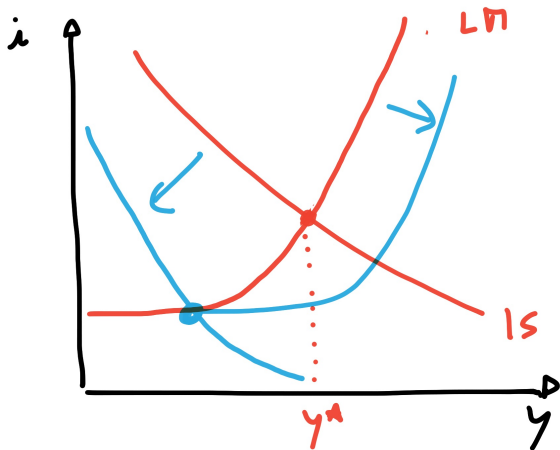
1. IS-LM

Figure 7: The liquidity trap



1. IS-LM

Figure 8: The liquidity trap



2. Facts

- ▶ We look at facts to show that the liquidity trap is not a theoretical curiosity:
 - × the US in the 1930's (Great Depression),
 - × Japan in the 1990's (Lost Decade),
 - × the US and the UK after 2008 (Great Recession).

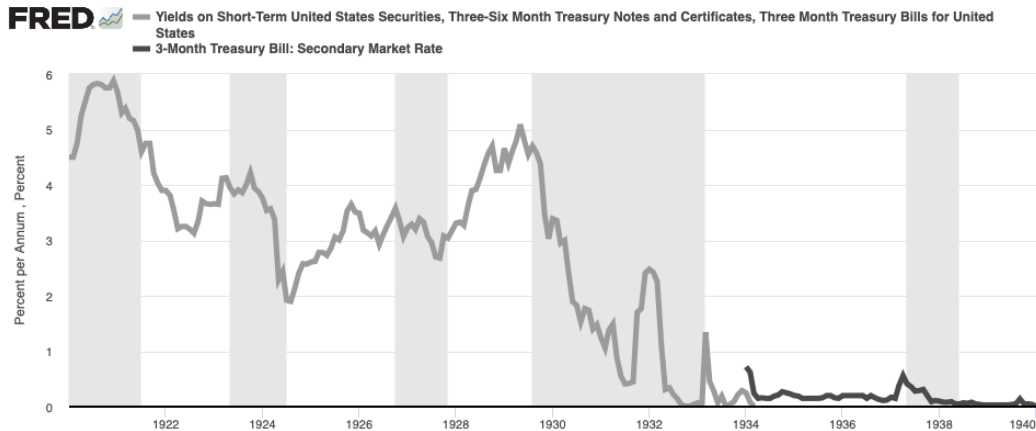
2. Facts

- ▶ In all cases, we observe a similar pattern:
 - × Asset prices (stock market) crashes
 - × nominal interest rate is almost zero
 - × there is a recession in output
 - × unemployment is high
 - × there is a initial period of deflation (not prolonged for the 2008 recession)

2. Facts

The US Great Depression

Figure 9: Short-term interest rates



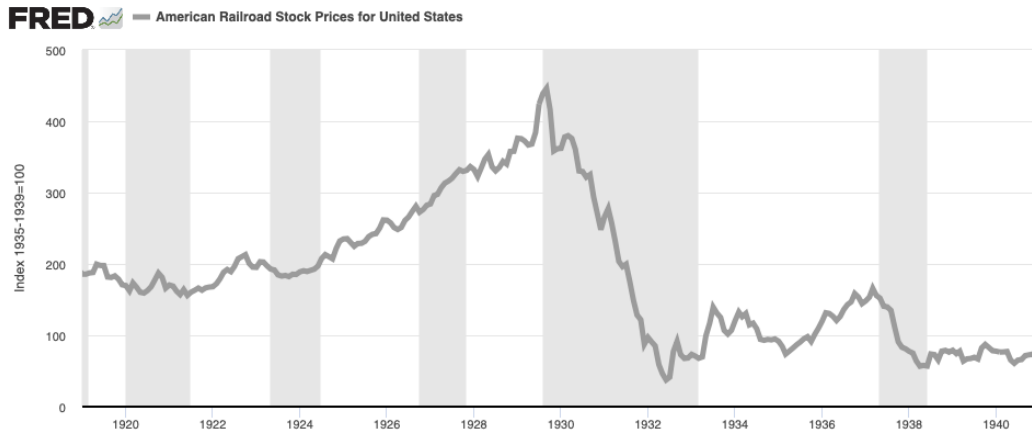
Sources: NBER; Board of Governors

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2. Facts

The US Great Depression

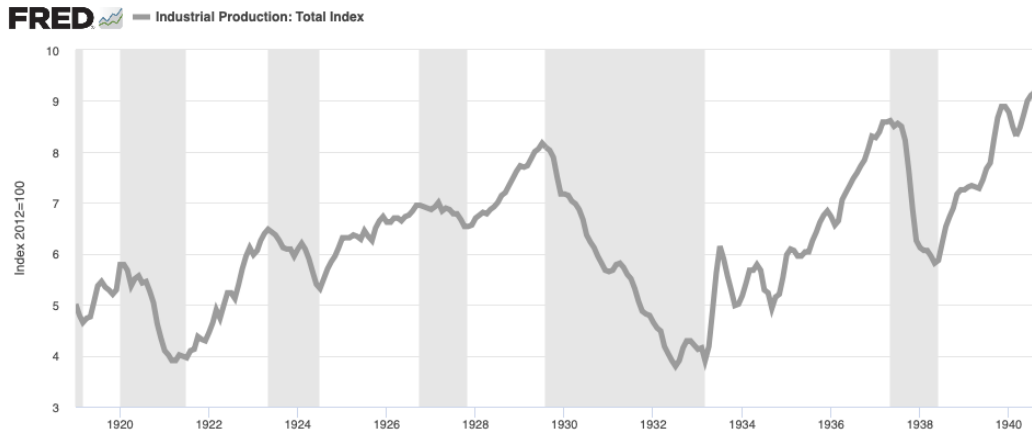
Figure 10: Equity prices



2. Facts

The US Great Depression

Figure 11: Industrial production



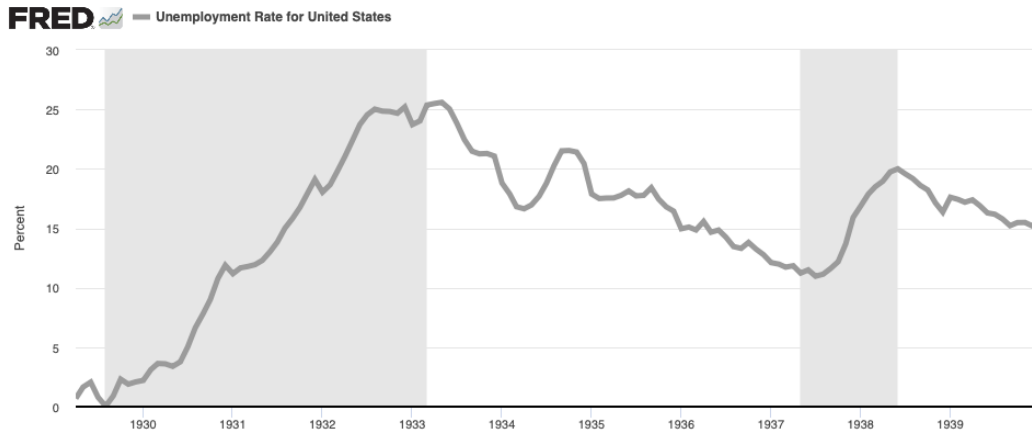
Source: Board of Governors of the Federal Reserve System (US)

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2. Facts

The US Great Depression

Figure 12: Unemployment rate



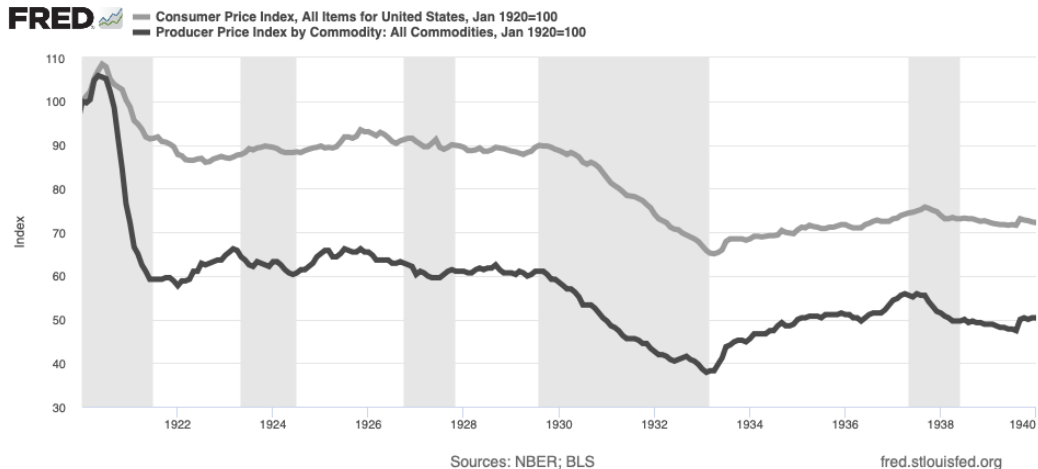
Source: National Bureau of Economic Research

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2. Facts

The US Great Depression

Figure 13: Consumer and Producer Prices



2. Facts

The Japan's lost decade of the 1990s

Figure 14: Discount rate



Source: International Monetary Fund

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2. Facts

The Japan's lost decade of the 1990s

Figure 15: Equity prices (Nikkei Index)



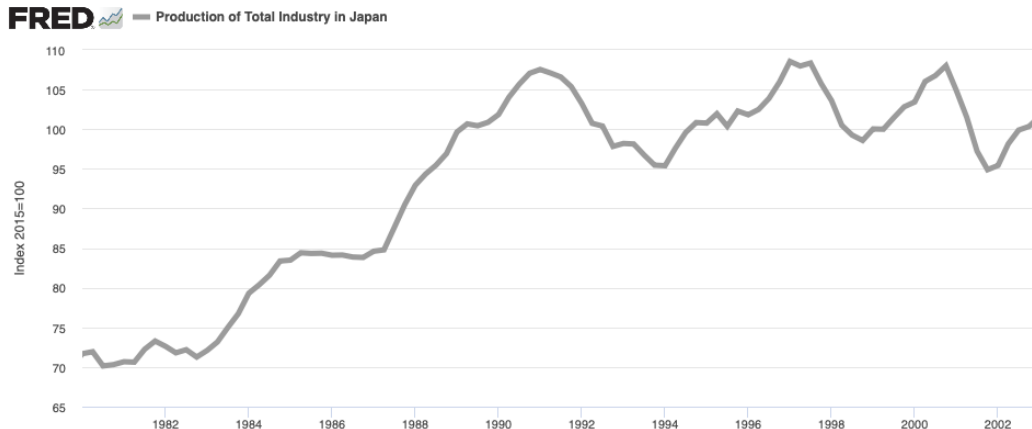
Source: Nikkei Industry Research Institute

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2. Facts

The Japan's lost decade of the 1990s

Figure 16: Industrial production



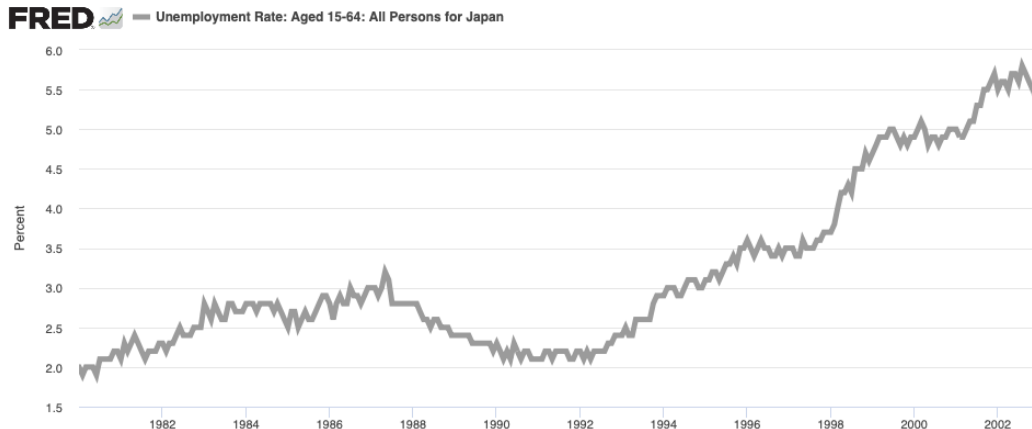
Source: Organization for Economic Co-operation and Development

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2. Facts

The Japan's lost decade of the 1990s

Figure 17: Unemployment rate



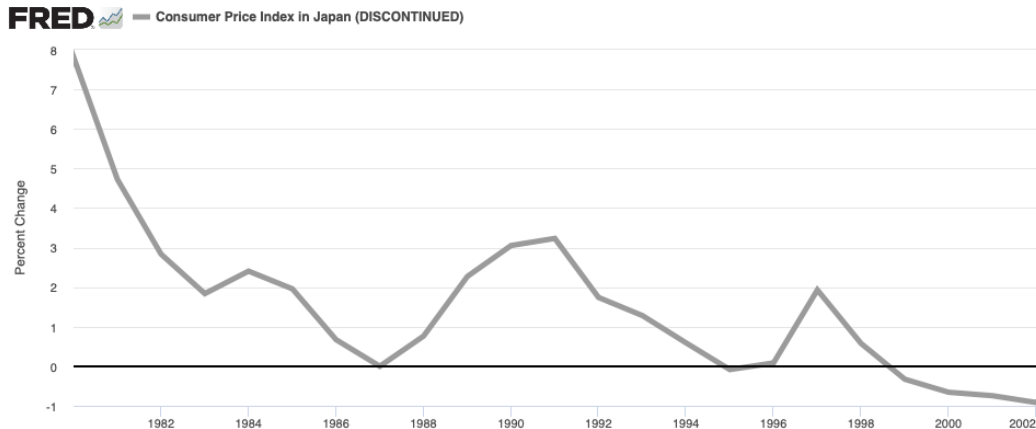
Source: Organization for Economic Co-operation and Development

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2. Facts

The Japan's lost decade of the 1990s

Figure 18: Inflation (CPI and GDP deflator)



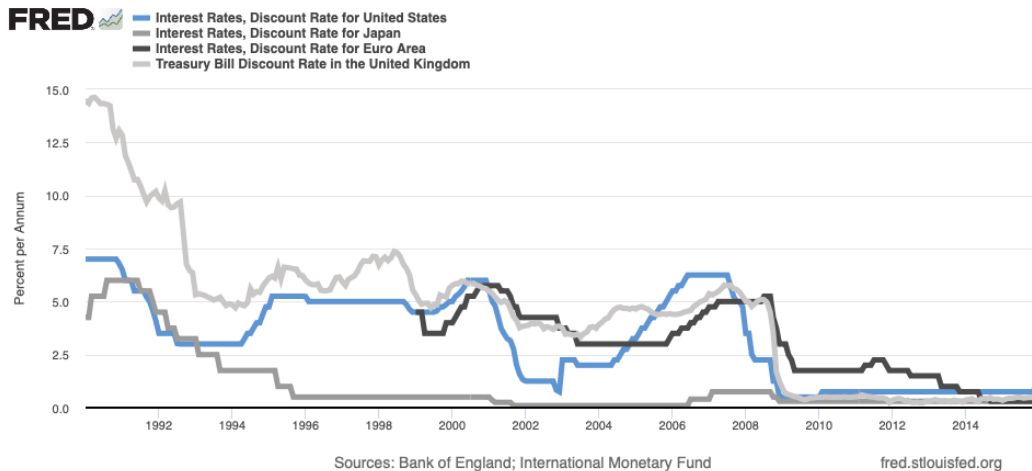
Source: U.S. Bureau of Labor Statistics

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2. Facts

The Great Recession

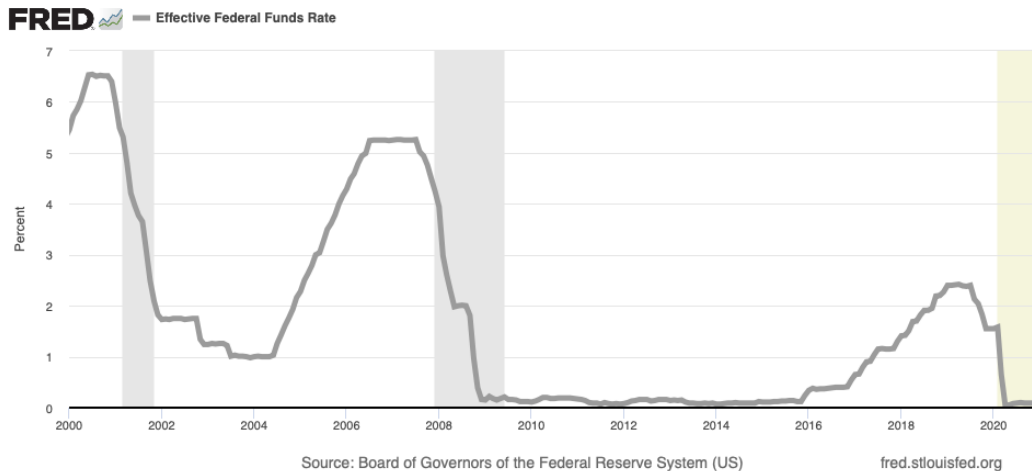
Figure 19: Short term interest rates



2. Facts

The US Great Recession

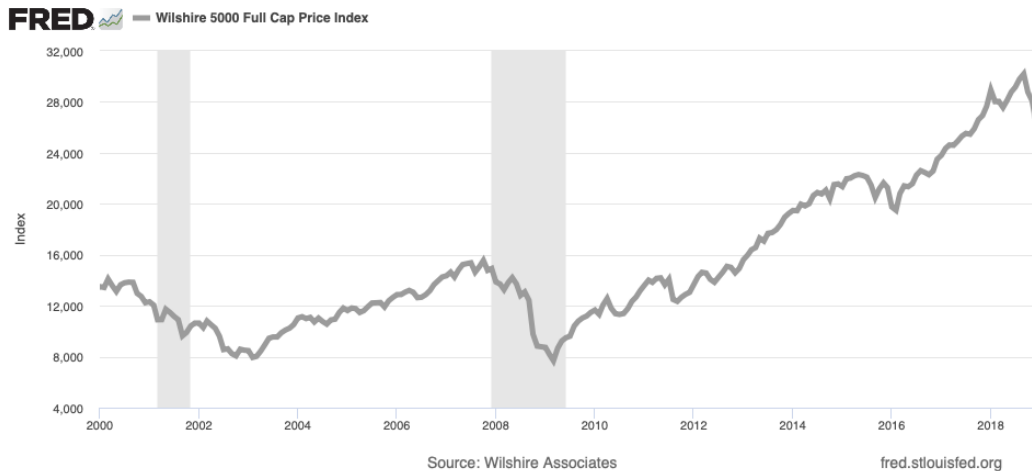
Figure 20: Federal Fund Rate



2. Facts

The US Great Recession

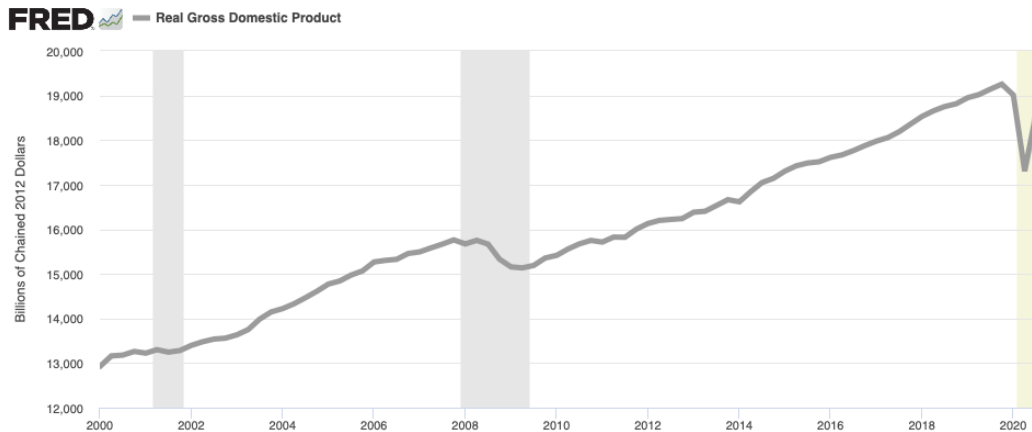
Figure 21: Equity prices



2. Facts

The US Great Recession

Figure 22: Real GDP



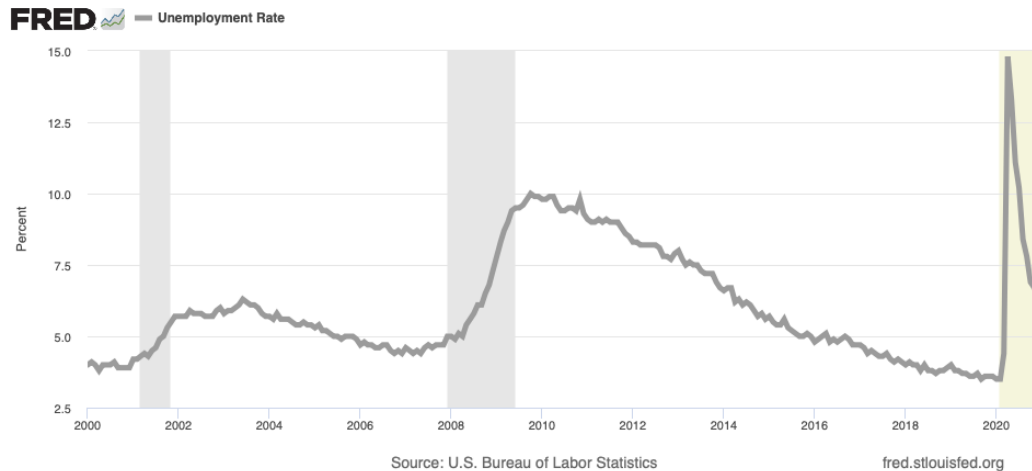
Source: U.S. Bureau of Economic Analysis

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2. Facts

The US Great Recession

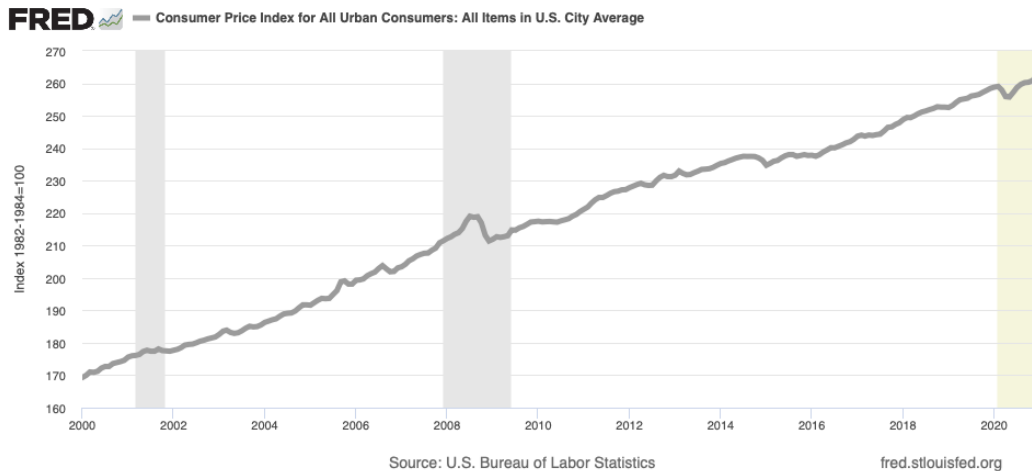
Figure 23: Unemployment rate



2. Facts

The US Great Recession

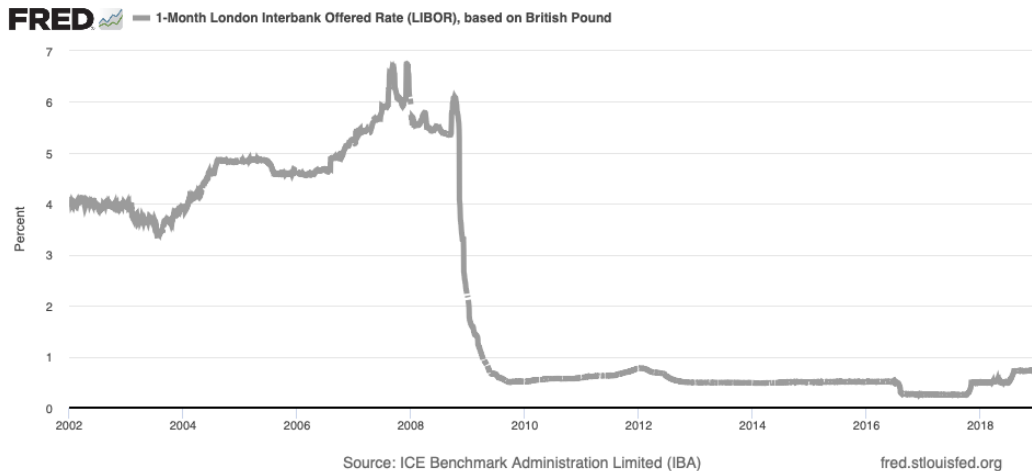
Figure 24: Consumer Price Index)



2. Facts

The UK Great Recession

Figure 25: 90-day Interbank Rate



2. Facts

The UK Great Recession

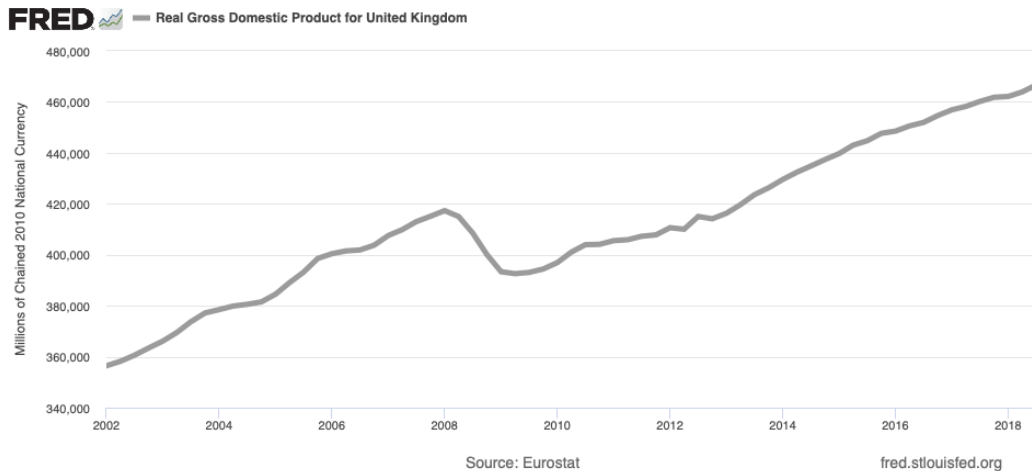
Figure 26: Equity prices (Total Share Price Index)



2. Facts

The UK Great Recession

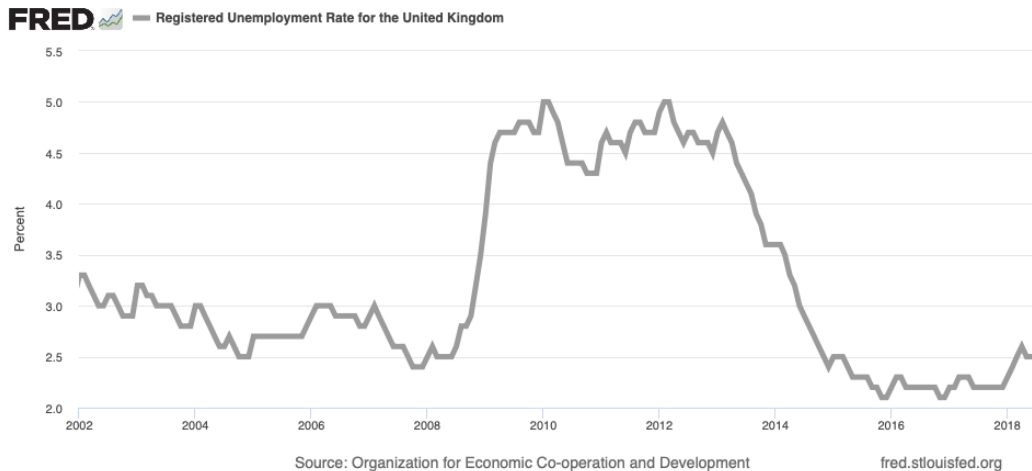
Figure 27: Real GDP



2. Facts

The UK Great Recession

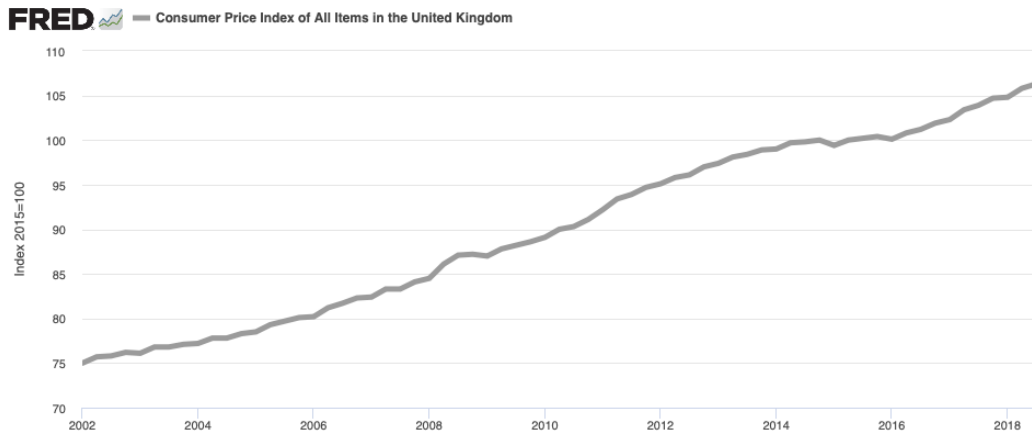
Figure 28: Unemployment rate



2. Facts

The UK Great Recession

Figure 29: Consumer Price Index



Source: Organization for Economic Co-operation and Development

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3. A Microfounded Model

The model

- ▶ A model proposed by PAUL KRUGMAN (1998)
- ▶ It gives a general equilibrium with micro-foundations and dynamics to IS-LM
- ▶ Deterministic closed economy
- ▶ One representative household

$$U = \sum_{t=0}^{\infty} \beta^t u(c_t)$$

with $1 > \beta > 0$ and $u'(c_t) > 0$.

- ▶ Endowment economy: y_t given and perishable
- ▶ Two assets : Money M_t and Bonds B_t
- ▶ Taboo: one cannot consume their own endowment (must sell it and buy some from someone else)
- ▶ “Others” consume $\bar{c}_t$.

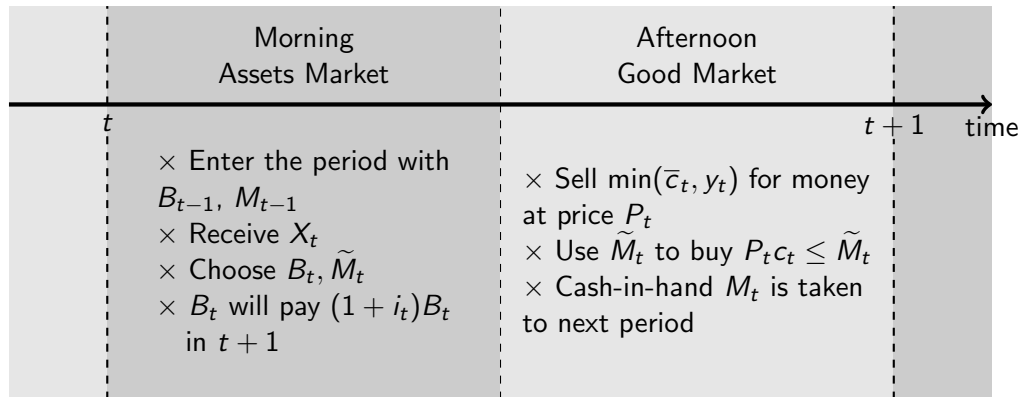
3. A Microfounded Model

The model

- ▶ Cash in Advance constraint

$$P_t c_t \leq \tilde{M}_t$$

- ▶ Timeline:



3. A Microfounded Model

Solving for the household optimal behavior

- The Hh maximizes utility under a sequence of period t budget constraint and cash in advance constraint:

Morning: Asset market

$$B_t + \tilde{M}_t \leq (1 + i_{t-1})B_{t-1} + M_{t-1} + X_t \quad (\lambda_t)$$

Afternoon: Good market

$$P_t c_t \leq \tilde{M}_t \quad (\mu_t)$$

$$M_t \leq \left(\tilde{M}_t - P_t c_t \right) + P_t \bar{c}_t \quad (\nu_t)$$

- X_t is newly created money distributed to Hh
- λ_t , μ_t and ν_t are positive or nul Lagrange multipliers.

3. A Microfounded Model

Solving for the household optimal behavior

- The Lagrangian is

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \beta^t \bigg\{ & u(c_t) \\ & + \lambda_t \left((1 + i_{t-1})B_{t-1} + M_{t-1} + X_t - B_t - \tilde{M}_t \right) \\ & + \mu_t \left(\tilde{M}_t - P_t c_t \right) \\ & + \nu_t \left(\tilde{M}_t - P_t c_t + P_t \bar{c}_t - M_t \right) \bigg\} \end{aligned}$$

- with $M_{-1} = M$ and $B_{-1} = 0$ given.
- $\mathcal{L}$ is maximized by choosing optimally $\{c_t, B_t, M_t, \tilde{M}_t\}_{t=0}^{\infty}$.

3. A Microfounded Model

Solving for the household optimal behavior

► and FOC are $\forall t \geq 0$

$$\text{w.r.t. } c_t : u'(c_t) = P_t(\mu_t + \nu_t) \quad (1)$$

$$\text{w.r.t. } B_t : \lambda_t = \beta(1 + i_t)\lambda_{t+1} \quad (2)$$

$$\text{w.r.t. } M_t : \nu_t = \beta\lambda_{t+1} \quad (3)$$

$$\text{w.r.t. } \tilde{M}_t : \lambda_t = \mu_t + \nu_t \quad (4)$$

$$\text{positivity} : \lambda_t \geq 0$$

$$\text{CS} : \lambda_t \left((1 + i_{t-1})B_{t-1} + M_{t-1} + X_t - B_t - \tilde{M}_t \right) = 0$$

$$\text{positivity} : \mu_t \geq 0$$

$$\text{CS} : \mu_t(\tilde{M}_t - P_t c_t) = 0$$

$$\text{positivity} : \nu_t \geq 0$$

$$\text{CS} : \nu_t(\tilde{M}_t - P_t c_t + P_t \bar{c}_t - M_t) = 0$$

(CS: *Complementary Slackness condition*)

3. A Microfounded Model

Solving for the household optimal behavior

Proposition 1

Under the assumption $u'(\cdot) > 0$, the morning and afternoon budget constraints always bind and $\lambda_t > 0$, $\nu_t > 0 \forall t \geq 0$,

Proof : From (1), $u'(c_t) > 0$ implies $\mu_t + \nu_t > 0$ (price $P_t > 0$). As (4) is $\lambda_t = \mu_t + \nu_t$, it follows that $\lambda_t > 0 \forall t$. Then (3) implies $\nu_t > 0 \forall t$. As the two multipliers are strictly positive, the complementary slackness conditions imply that the associated constraints are satisfied with an equality.

3. A Microfounded Model

Solving for the household optimal behavior

Proposition 2

If $i_t > 0$, then $\mu_t > 0$ and $P_t c_t = \tilde{M}_t$. If $i_t = 0$, then $\mu_t = 0$ and $P_t c_t \leq \tilde{M}_t$.

Proof : (3) and (4) imply $\mu_t = \lambda_t - \beta\lambda_{t+1}$. (2) rewrites $\lambda_t - \beta\lambda_{t+1} = \beta\lambda_{t+1}i_t$. We therefore have $\mu_t = \beta\lambda_{t+1}i_t$. From this last equation, we have the first part of the proposition.

Then from the complementary slackness condition $\mu_t(\tilde{M}_t - P_t c_t) = 0$, we obtain:

- If $i_t > 0$, then the CIA constraint binds:

$$P_t c_t = \tilde{M}_t$$

- If $i_t = 0$, the CIA constraint may not be binding : Hh is indifferent between money and bonds.

3. A Microfounded Model

Equilibrium with flexible prices

Definition 1

A symmetric competitive equilibrium with flexible prices is a sequence of strictly positive prices and positive interest rates $\{P_t, i_t\}_{t=0}^{\infty}$ and positive quantities $\{\tilde{M}_t, M_t, c_t, \bar{c}_t, B_t\}_{t=0}^{\infty}$ such that, given monetary injections $\{X_t\}_{t=0}^{\infty}$, endowments $\{y_t\}_{t=0}^{\infty}$ and for $M_{-1} = M$ and $B_{-1} = 0$ given:

- ▶ *Given prices, quantities solve the Hh maximisation problem,*
- ▶ *Good, money and bond markets clear and $c_t = \bar{c}_t$ (symmetry)*
- ▶ *$P_t > 0, i_t \geq 0, \tilde{M}_t \geq 0, M_t \geq 0, c_t \geq 0, \bar{c}_t \geq 0$.*

3. A Microfounded Model

Equilibrium with flexible prices

- ▶ All agents are identical (representative household, therefore symmetric equilibrium):

$$\bar{c}_t = c_t.$$

- ▶ The equilibrium on the bond market is

$$B_t = 0.$$

- ▶ Money market equilibrium:

$$M_t = \tilde{M}_t = M_{t-1} + X_t.$$

- ▶ Good market equilibrium

$$c_t = y_t.$$

- ▶ Remark : the equilibrium quantities can be determined without using the FOC
- ▶ The FOC are determining *equilibrium prices*, i.e. the prices P and i such that markets clear.

3. A Microfounded Model

Equilibrium with flexible prices

- ▶ In what follows, we restrict to a specific case:
 - × Period 0: $y_0 = y$, $M_0 = M + X_0$, $X_0 \neq 0$ if money creation.
 - × Period 1: $y_1 = y$, $M_1 = M \Leftrightarrow X_1 = -X_0$.
 - × for $t \geq 1$, $y_t = y$, $M_t = M$, $X_t = 0$.
- ▶ When money is created in period 0, it is only for one period $\rightsquigarrow$ it is destroyed in period 1. (not the usual assumption)
- ▶ We can then solve the model in the following way
 - × Solve for period 1 and onwards
 - × Solve for period 0

3. A Microfounded Model

Equilibrium with flexible prices - Period 1 and onwards

- ▶ All variables are constant.
- ▶ $c = y$
- ▶ (2) writes $(1 + i) = 1/\beta$ so that

$$i = \frac{1}{\beta} - 1 > 0.$$

Therefore the CIA constraint binds and $P = M/y$.

- ▶ Assume for analytical tractability that $u(c) = \log c$
- ▶ $\lambda = \frac{1}{M}$ and $\mu = \frac{(1-\beta)}{M}$, $\nu = \frac{\beta}{M}$.

3. A Microfounded Model

Computation of λ , μ and ν

- ▶ The CIA constraint binds because $i > 0$.
- ▶ (1) writes $\mu + \nu = \frac{1}{P_c}$ which gives using the CIA constraint $\mu + \nu = \frac{1}{M}$. From (4), we obtain $\lambda = \frac{1}{M}$.
- ▶ (3) writes $\nu = \beta\lambda = \frac{\beta}{M}$.
- ▶ As $\mu + \nu = \frac{1}{M}$, $\mu = \frac{(1-\beta)}{M}$.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

- ▶ The equilibrium in period $t \geq 1$ is $c = y$, $i = 1/\beta - 1$, $P = M/y$
- ▶ The equilibrium (i_0, P_0) in period 0 is given by the equations: (for (a), we use $\lambda_0 = \mu_0 + \nu_0$ to rewrite (1) as (a)) (we rewrite (2) using (3) and (4) as (c)) (we also make use of $c_0 = y_0$ (= y))

$$\frac{1}{y_0} = P_0 \lambda_0 \quad (a)$$

$$\lambda_0 = \beta(1 + i_0)\lambda \quad (b)$$

$$\mu_0 = \beta \lambda i_0 \quad (c)$$

$$\mu_0(M_0 - P_0 y_0) = 0 \quad (d)$$

- ▶ 4 equations with 4 unknowns $(i_0, P_0, \lambda_0, \mu_0)$.
- ▶ Let's get rid of λ_0 and μ_0 .

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

- (a) and (b) imply, using the value of λ :

$$i_0 = \frac{1}{\beta} \frac{Py}{P_0 y_0} - 1. \quad (A)$$

- Finally, (c) and (d) imply

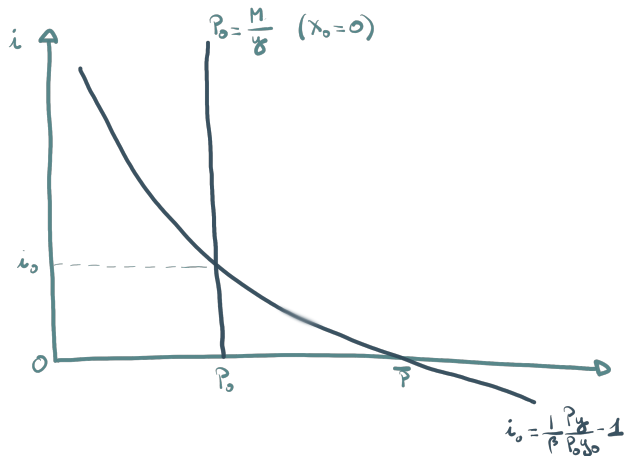
$$P_0 = \frac{M_0}{y_0} \text{ if } i_0 > 0. \quad (B)$$

- Using (A), $i_0 > 0$ implies $P_0 < \bar{P} = \frac{Py}{\beta y_0} = \frac{1}{\beta} \frac{M}{\beta y_0}$ (using $P = \frac{M}{y}$)
- Note that if $X_0 = 0$ and $y_0 = y$, $P_0 = P < \bar{P}$.
- In the following, we assume $y_0 = y$.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

Figure 30: Equilibrium in period 0 - Flexible prices



3. A Microfounded Model

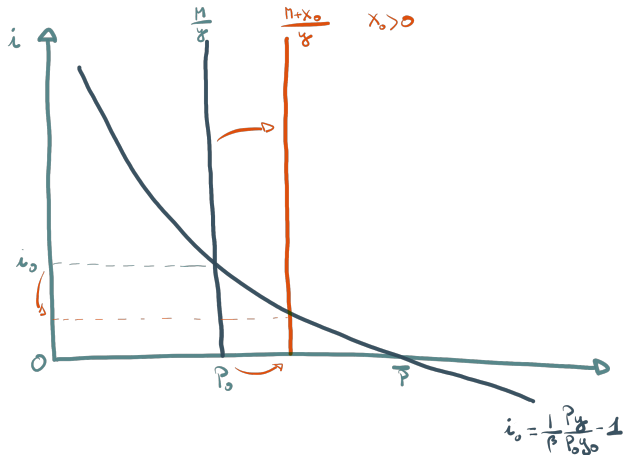
Equilibrium with flexible prices - Period 0

- ▶ How does the model works?
- ▶ Suppose an increase in money supply today $X_0 > 0$ (not too large)
- ▶ Prices today P_0 increase
- ▶ As P is given, inflation between and tomorrow is reduced (there is deflation as $P_1 = P < P_0$).
- ▶ Consumption is independent of monetary policy, $c_0 = y_0 = y$ (money is neutral).
- ▶ Therefore, the decrease in inflation decreases i_0 .
- ▶ Important : the above analysis is true as long as $P_0 < \bar{P}$, such that $i_0 > 0$ and the CIA binds.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0

Figure 31: Equilibrium in period 0 - Flexible prices - X_0 not too large

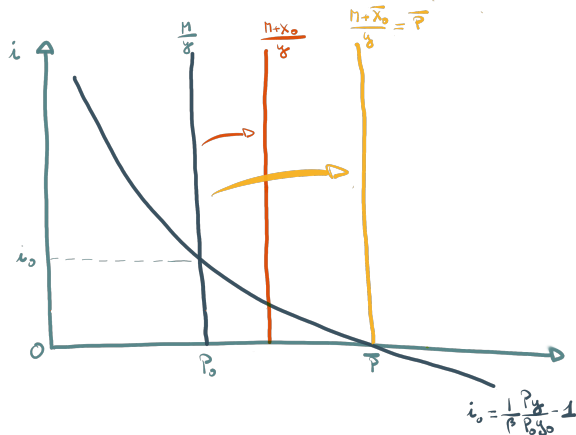


3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

- Consider the money creation $\bar{X}$ such that $P_0 = \bar{P}$ and $i_0 = 0$

Figure 32: Equilibrium in period 0 - Flexible prices - $X_0 = \bar{X}_0$



3. A Microfounded Model

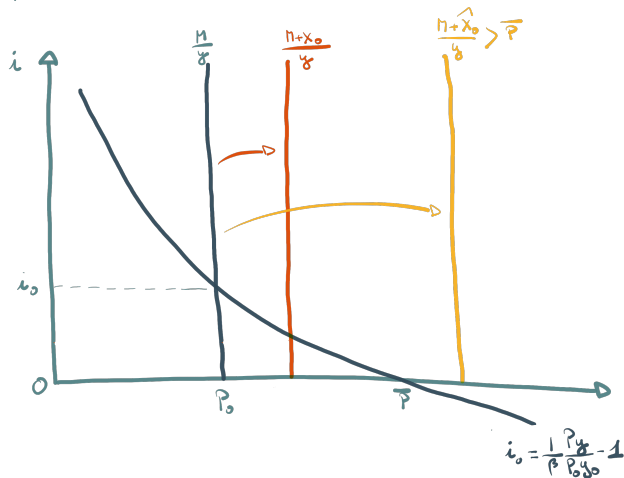
Equilibrium with flexible prices - Period 0 - The liquidity trap

- ▶ Consider the money creation $\bar{X}_0$ such that $P_0 = \bar{P}$ and $i_0 = 0$
- ▶ What does happen if money creation is $\tilde{X} > \bar{X}$?
- ▶ If P_0 increases above $\bar{P}$, then $i_0 < 0$ which is not possible.
- ▶ Therefore, we must have $i_0 = 0$, and then $P_0 = \bar{P}$
- ▶ As $i_0 = 0$, the CIA does not bind
- ▶ A decrease in i is not needed to absorb the extra money creation $\rightsquigarrow$ “liquidity trap”.

3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

Figure 33: Equilibrium in period 0 – Flexible prices – X_0 large



3. A Microfounded Model

Equilibrium with flexible prices - Period 0 - The liquidity trap

- ▶ Remark : In the liquidity trap with flexible prices, money cannot reduce inflation below $\frac{P}{P}$ (which is actually *deflation*)
- ▶ But money is always neutral ... so that we do not care much about the liquidity trap.
- ▶ Things will be different if we assume, as in IS-LM, that prices are fixed in period 0.

3. A Microfounded Model

Equilibrium with fix prices - Period 0

- ▶ Now P_0 is fixed (given exogenously), and assume $P_0 > P$ and $X_0 = 0$ (so that $M_0 = M$).
- ▶ As $M_0 = M < P_0 y$, households can only buy $c_0 = \frac{M}{P_0} < y$.
- ▶ Period-0 allocations are still given by the four equations

$$\frac{1}{c_0} = P_0 \lambda_0 \quad (a)$$

$$\lambda_0 = \beta(1 + i_0)\lambda \quad (b)$$

$$\mu_0 = \beta \lambda i_0 \quad (c)$$

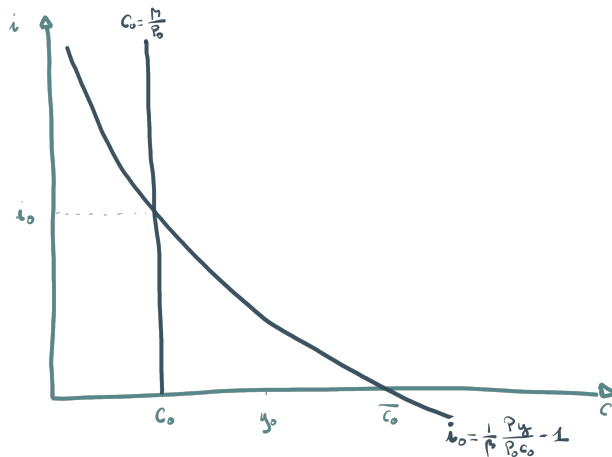
$$\mu_0(M_0 - P_0 c_0) = 0 \quad (d)$$

- ▶ 4 equations with 4 unknowns ($i_0, c_0, \lambda_0, \mu_0$).
- ▶ For $y_0 = y$, $P_0 > P$ makes the CIA constraint binding (see next figure), and we have $P_0 c_0 = M_0$, with $c_0 \leq y_0$
- ▶ Monetary policy now determines c_0 as P_0 is fixed.

3. A Microfounded Model

Equilibrium with fix prices - Period 0

Figure 34: Equilibrium in period 0 - Fix prices



3. A Microfounded Model

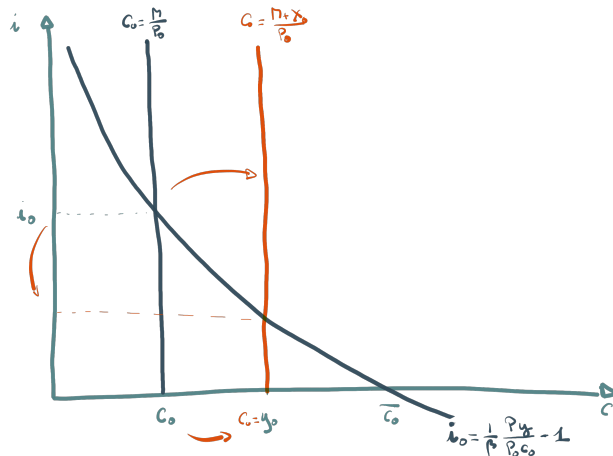
Equilibrium with fix prices - Period 0

- ▶ The economy is in a configuration similar to IS-LM
- ▶ As $c_0 < y_0$, the equilibrium is inefficient (an amount $y_0 - c_0$ of goods is wasted)
- ▶ Assume money creation $X_0 > 0$: to clear the money market, i_0 needs to go down. As inflation is fixed, this is achieved by an increase in $c_0 \rightsquigarrow$ expansionary monetary policy increases c_0 .
- ▶ Money is not neutral.

3. A Microfounded Model

Equilibrium with fix prices - Period 0

Figure 35: Equilibrium in period 0 - Fix prices - Monetary policy is *effective*



3. A Microfounded Model

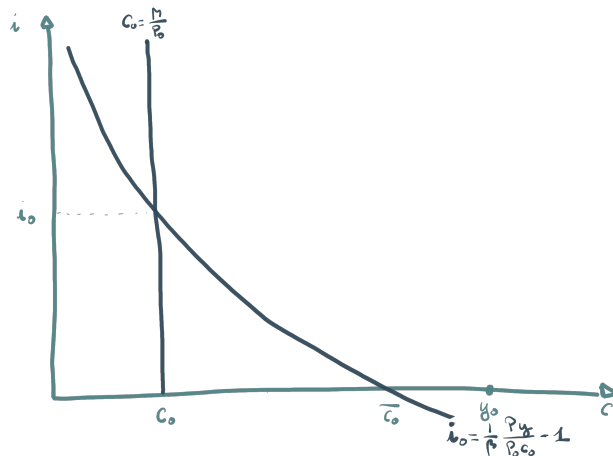
Equilibrium with fix prices - Period 0 - The liquidity trap

- ▶ Now assume that P_0 is large, such that $\bar{c}_0 < y_0$.
- ▶ $\bar{c}_0$ is the level of consumption for which, given P_0 , the nominal interest rate i_0 is zero.

3. A Microfounded Model

Equilibrium with fix prices - Period 0 - The liquidity trap

Figure 36: Equilibrium in period 0 - Fix prices



3. A Microfounded Model

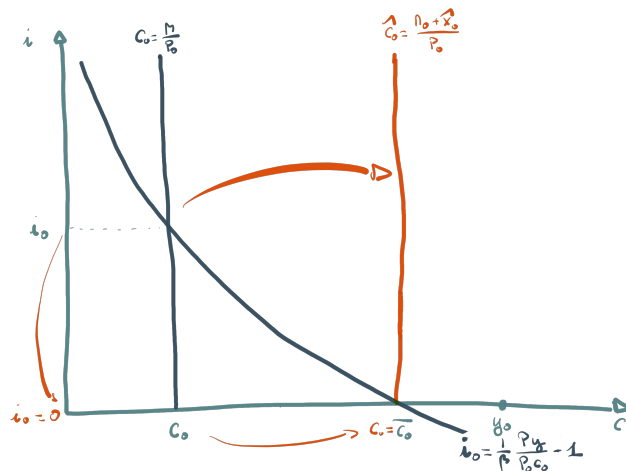
Equilibrium with fix prices - Period 0 - The liquidity trap

- ▶ Let's do a "large" money injection $\hat{X}_0$.
- ▶ The economy enters a liquidity trap
- ▶ The CIA is not binding anymore: the money increases will not decrease i_0
- ▶ The economy is stuck at $c_0 = \bar{c}_0, y_0$ and monetary policy is ineffective in increasing further c_0 .
- ▶ For a money creation $\tilde{X}_0 > \hat{X}_0$, no further increase in c_0

3. A Microfounded Model

Equilibrium with fix prices - Period 0 - The liquidity trap

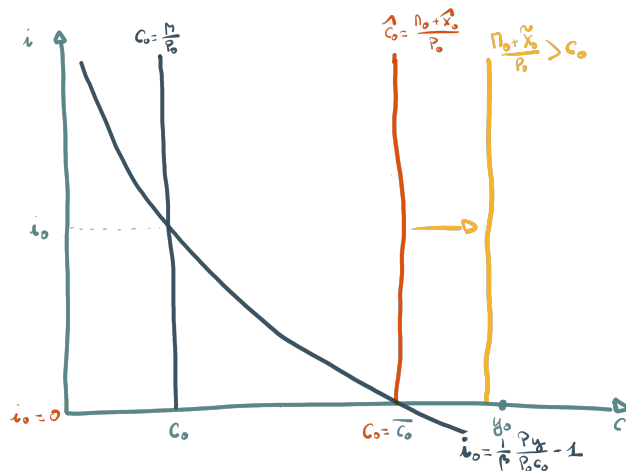
Figure 37: Equilibrium in period 0 - Fix prices



3. A Microfounded Model

Equilibrium with fix prices - Period 0 - The liquidity trap - Monetary policy is ineffective

Figure 38: Equilibrium in period 0 - Fix prices



4. Effective policy in the liquidity trap

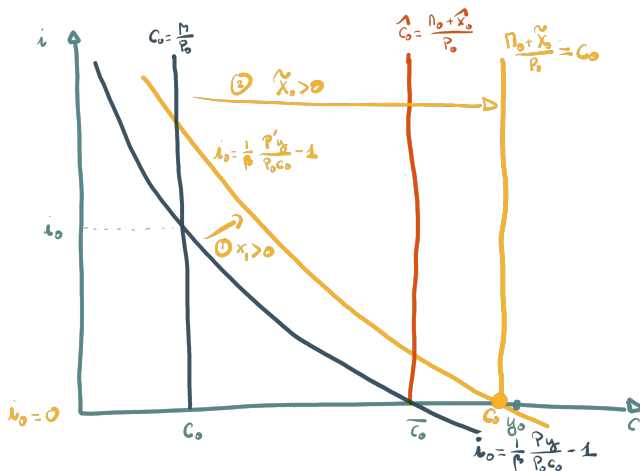
Manipulating expectations: “committing to be irresponsible”

- ▶ If the Central Bank can raise the inflation expectations of the agents (P/P_0) by being credible to create money tomorrow:
- ▶ $X_1 > 0$ so that $M_t = M + X_1 \forall t \geq 1$.
- ▶ The new price level for $t \geq 1$, P' , will be now higher.
- ▶ Then there will be an increase in the nominal interest rate today i_0 ,
- ▶ and it is therefore possible to create money today $\tilde{X}_0$ and narrow the gap between c_0 and y_0 .

4. Effective policy in the liquidity trap

Manipulating expectations: "committing to be irresponsible"

Figure 39: Equilibrium in period 0 - Fix prices - increasing M and committing to increase M^*



5. References

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- ▶ P. Krugman, 1998, "*It's Baaack: Japan's Slump and the Return of the Liquidity Trap*", *Brookings Papers on Economic Activity*, Vol. 29
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